

Лабораторная работа

Номер 5

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Информация

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Приобретение практических навыков по расширенному конфигурированию HTTP-сервера Apache в части безопасности и возможности использования PHP.

Запуск сервера

```
C:\work\nsandyushin\vagrant>vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Clearing any previously set forwarded ports...
==> server: Clearing any previously set network interfaces...
==> server: Preparing network interfaces based on configuration...
    server: Adapter 1: nat
    server: Adapter 2: intnet
==> server: Forwarding ports...
    server: 22 (guest) => 2222 (host) (adapter 1)
==> server: Running 'pre-boot' VM customizations...
==> server: Booting VM...
==> server: Waiting for machine to boot. This may take a few minutes...
    server: SSH address: 127.0.0.1:2222
    server: SSH username: vagrant
    server: SSH auth method: password
==> server: Machine booted and ready!
[server] GuestAdditions 7.1.4 running --- OK.
==> server: Checking for guest additions in VM...
==> server: Setting hostname...
==> server: Configuring and enabling network interfaces...
==> server: Mounting shared folders...
    server: C:/work/nsandyushin/vagrant => /vagrant
==> server: Machine already provisioned. Run 'vagrant provision' or use the '--provision'
==> server: flag to force provisioning. Provisioners marked to run always will still run.
==> server: Running provisioner: common hostname (shell)...
    server: Running: C:/Users/mega_/AppData/Local/Temp/vagrant-shell20251004-14476-fxwvc5.sh
```

Рис. 1: Запуск сервера

Создание сертификата

[illegible]

Рис. 2: Создание сертификата

Редактирование конфигурационного файла httpd

```
[root@server.nsandryushin.net private]# cd /etc/httpd/conf.d  
[root@server.nsandryushin.net conf.d]# nano www.nsandryushin.net.conf
```

Рис. 3: Редактирование конфигурационного файла httpd

```
GNU nano 8.1                               www.nsandryushin.net.conf
<VirtualHost *:80>
    ServerAdmin webmaster@nsandryushin.net
    DocumentRoot /var/www/html/www.nsandryushin.net
    ServerName www.nsandryushin.net
    ServerAlias www.nsandryushin.net
    ErrorLog logs/www.nsandryushin.net-error_log
    CustomLog logs/www.nsandryushin.net-access_log common
    RewriteEngine on
    RewriteRule ^(.*)$ https://%{HTTP_HOST}$1 [R=301,L]
</VirtualHost>

<IfModule mod_ssl.c>
<VirtualHost *:443>
    SSLEngine on
    ServerAdmin webmaster@nsandryushin.net
    DocumentRoot /var/www/html/www.nsandryushin.net
    ServerName www.nsandryushin.net
    ServerAlias www.nsandryushin.net
    ErrorLog logs/www.nsandryushin.net-error_log
    CustomLog logs/www.nsandryushin.net-access_log common
    SSLCertificateFile /etc/ssl/certs/www.nsandryushin.net.crt
    SSLCertificateKeyFile /etc/ssl/private/www.nsandryushin.net.key
</VirtualHost>
</IfModule>
```

Рис. 4: www.nsandryushin.net.conf

Настройка фаервола

```
[root@server.nsandiyushin.net conf.d]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http ssh
[root@server.nsandiyushin.net conf.d]# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client aosp an
aps anno-1502 anno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-dire
ctor bareos-filerdemon bareos-storage bb bbg bitcoind bitcoind-rpc bitcoind-testnet bitcoind-t
estnet-ipc bitcoind-torrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent civilization-
tv civilization-v cockpit collectd condor-collector cratedb ctdb dds dds-multicast dds-unl
cast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-qluc dns-over-tls docker-registry docke
r-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger foreman fore
man-proxy freepa-4 freepa-ldap freepa-ldaps freepa-replication freepa-trust ftp galer
a ganglia-client ganglia-master git gssd grafana gre high-availability http http3 https id
ent ldap lnaps lperf2 lperf3 ipfs ipn ipn-client ipsec irc ircs iscsi-target isns jenkins
kadm5 kdeconnect kerberos kibana klogind kpasswd kprop kshell kube-api kube-apiserver kube
-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-s
ecure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kube
let-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llnet llnet-c
lient llnet-top llnet-udp managesieve matrix mdns memcache minecraft minidlna mdp mongod
mosh mountd epd mqtt mqtt-tls ns-wbt mysql mzymur mysql nbd nebula need-for-speed-most-wa
nted netbios-ns netdata-dashboards nfs nfs3 nmap-0183 nrip ntp nut opentelemetry openvpn ov
irt-lmaoio ovirt-storageconsole ovirt-vmconsole plex pexd pmpoxy pwebapi pwebapis pop3
pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netstv
ptp pulseaudio puppetmaster quassel radius radsec rdp redis redis-sentinel rootd rpc-bind
rquotad rsh rsyncd itss salt-master samba samba-client samba-dc sane settlers-history-col
lection sip slips slimevr slp setp smtp-submission smtps snmp snmptls snmptls-trap snmptrap
spideroak-lansync spotify-sync squid sssd ssh statvs steam-lan-transfer steam-streaming
stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn syncthing synct
hing-gui syncthing-relay synergy sysconlan syslog syslog-tls telnet tentacle terraria tftp
tile38 tinc tor-socks transmission-client turn turns upnp-client vds vnc-server vrrp war
pinator wben-http wben-https wireguard ws-discovery ws-discovery-client ws-discovery-host
ws-discovery-top ws-discovery-udp wssd wssd-http wspan wspan-xmpp xmpp-xmpp-client
xmpp-local xmpp-server zabbix-agent zabbix-java-gateway zabbix-server zabbix-trapper zabbix
x-web-service zero-k zerotier
[root@server.nsandiyushin.net conf.d]# firewall-cmd --add-service=https
success
[root@server.nsandiyushin.net conf.d]# firewall-cmd --add-service=https --permanent
success
[root@server.nsandiyushin.net conf.d]# firewall-cmd --reload
success
[root@server.nsandiyushin.net conf.d]# systemctl restart httpd
```

Рис. 5: Настройка фаервола

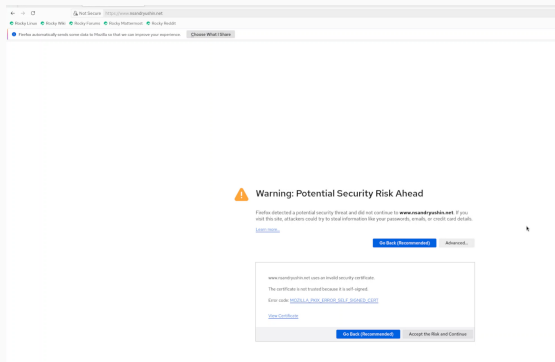


Рис. 6: Предупреждение

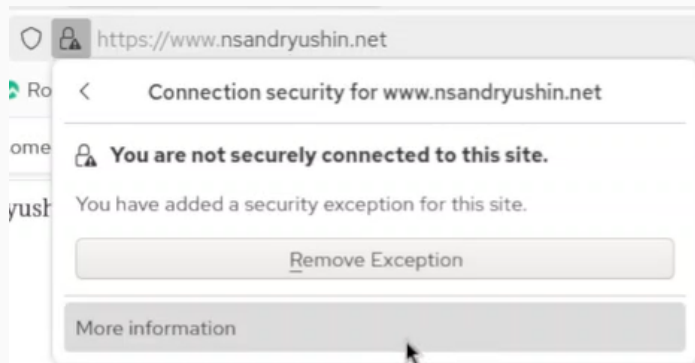


Рис. 7: Подключение к сайту

Установка php

```
[root@server.nsandryushin.net conf.d]# dnf -y install php
Last metadata expiration check: 0:10:57 ago on Sat 04 Oct 2025 04:52:42 PM UTC.
Dependencies resolved.
=====
Package                Architecture  Version           Repository         Size
=====
Installing:
php                    x86_64        8.3.19-1.el10_0   appstream          11 k
Installing dependencies:
capstone               x86_64        5.0.1-6.el10      appstream          1.0 M
nginxfilesystem        noarch        2:1.26.3-1.el10   appstream          11 k
php-common             x86_64        8.3.19-1.el10_0   appstream          713 k
Installing weak dependencies:
php-cli               x86_64        8.3.19-1.el10_0   appstream          3.6 M
php-fpm               x86_64        8.3.19-1.el10_0   appstream          1.9 M
php-mbstring          x86_64        8.3.19-1.el10_0   appstream          521 k
php-opcache           x86_64        8.3.19-1.el10_0   appstream          367 k
php-pdo               x86_64        8.3.19-1.el10_0   appstream          88 k
php-xml               x86_64        8.3.19-1.el10_0   appstream          149 k

Transaction Summary
=====
Install 10 Packages
```

Рис. 9: Установка php

Замена файла index.html на index.php

```
[root@server.nsandryushin.net conf.d]# touch /var/www/html/www.nsandryushin.net/index.php
[root@server.nsandryushin.net conf.d]# nano /var/www/html/www.nsandryushin.net/index.php
[root@server.nsandryushin.net conf.d]# rm /var/www/html/www.nsandryushin.net/index.html
rm: remove regular file '/var/www/html/www.nsandryushin.net/index.html'? y
[root@server.nsandryushin.net conf.d]#
```

Рис. 10: Замена файла index.html на index.php

A screenshot of a GNU nano 8.1 text editor window. The title bar at the top shows the file path /var/www/html/www.nsandryushin.net/index.php. The editor area contains three lines of PHP code: the opening tag <?php in green, the function call phpinfo(); in grey, and the closing tag ?> in green. A black cursor is positioned at the end of the third line.

```
GNU nano 8.1 /var/www/html/www.nsandryushin.net/index.php
<?php
phpinfo();
?>
```

Рис. 11: Содержимое index.php

Исправление настроек доступа

```
[root@server.nsandryushin.net conf.d]# chown -R apache:apache /var/www
[root@server.nsandryushin.net conf.d]# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
[root@server.nsandryushin.net conf.d]# restorecon -vR /var/www
[root@server.nsandryushin.net conf.d]# systemctl restart httpd
[root@server.nsandryushin.net conf.d]#
```

Рис. 12: Исправление настроек доступа

Отображение сайта

[illegible]

Рис. 13: Отображение сайта

Сохранение конфигурации

```
[root@server.nsandryushin.net conf.d]# cd /vagrant/provision/server/http
[root@server.nsandryushin.net http]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/autoindex.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/fcgid.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/manual.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/README'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/server.nsandryushin.net.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/ssl.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/userdir.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/welcome.conf'? y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/www.nsandryushin.net.conf'? y
[root@server.nsandryushin.net http]# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
cp: overwrite '/vagrant/provision/server/http/var/www/html/server.nsandryushin.net/index.html'? y
[root@server.nsandryushin.net http]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/private
[root@server.nsandryushin.net http]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/certs
[root@server.nsandryushin.net http]# cp -R /etc/pki/tls/private/www.nsandryushin.net.key /vagrant/provision/server/http/etc/pki/tls/private
[root@server.nsandryushin.net http]# cp -R /etc/pki/tls/certs/www.nsandryushin.net.crt /vagrant/provision/server/http/etc/pki/tls/certs
```

Рис. 14: Сохранение конфигурации

```
GNU nano 8.1 /vagrant/provision/server/http.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
systemctl enable httpd
systemctl start httpd

echo "install php"
dnf -y install php
echo "configure firewall for https"
firewall-cmd --list-services
firewall-cmd --get-services
firewall-cmd --add-service=https
firewall-cmd --add-service=https --permanent
firewall-cmd --reload
echo "restart httpd"
systemctl restart httpd
```

Рис. 15: http.sh

В результате выполнения лабораторной работы были получены навыки по конфигурированию HTTP-сервера Apache и https